Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

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Abstract

High-frequency surprises identify monetary policy shocks cleanly, but their estimated effects on unemployment and inflation after 1994 are often weak or puzzling. We hold the surprise fixed and change only how it is propagated. When the surprise reaches the economy through a nominal short rate alone, the familiar puzzles appear. When it propagates through financial conditions as a whole, measured relative to the neutral rate r^*, a policy move that raises the two-year yield 1 percentage point raises unemployment by about 0.4 percentage point within a year, lowers the core price level by about 0.6 percent after two years, and lowers industrial production by about 1 percent. Financial conditions are essential for the unemployment response; measuring rates and financial conditions relative to r^* is essential for prices and output.

1 Introduction

The high-frequency identification of monetary policy shocks is one of the major empirical advances in macroeconomics. Changes in interest-rate futures and Treasury yields in narrow windows around FOMC announcements isolate news about policy from other news (Kuttner, 2001; Gürkaynak et al., 2005), and their effects on asset prices are estimated precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021). Using these surprises to learn about the effects of policy on output, unemployment and inflation has been harder. In samples that start in the 1990s, the estimated responses are often small, imprecise or of the wrong sign (Ramey, 2016). The literature has made major progress on this problem. One line of work shows that announcements also reveal the Federal Reserve's information about the economy, so that a surprise tightening partly signals good news, and separates the two components (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). Another shows that surprises are partly predictable from public data that predate the announcement, and that orthogonalizing them, and adding Chair speeches, restores conventional and larger responses (Bauer and Swanson, 2023a,b). Closest to us, Gertler and Karadi (2015) show that policy surprises work through credit costs, and Caldara and Herbst (2019) that omitting credit spreads, to which policy also reacts, attenuates the response of activity. The link between futures-based surprises and monthly VARs goes back to Faust et al. (2004).

This note focuses on the other half of the exercise: how the surprise is propagated to the economy. We hold the high-frequency shock fixed and ask how much of the post-1994 puzzle reflects two features of the transmission mechanism that standard specifications leave out. First, policy affects spending through the whole configuration of financial conditions—the cost of long-term credit, the value of wealth and the exchange rate (Boivin et al., 2010)—not through a single short rate; summarizing them in a financial conditions index is common in policy analysis (Hatzius

et al., 2010; Ajello et al., 2023). Second, what matters for spending and inflation is the level of real rates relative to the neutral rate r^*, which fell substantially from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017; Hamilton et al., 2016; Kiley, 2020) and shaped the trend in yields (Bauer and Rudebusch, 2020). Over the post-1994 period, rates trended down with r^* while activity trended up; a reduced-form link between the level of rates and activity then partly reflects these opposing trends rather than the cyclical stance of policy.

Our design is direct. The shock is the Nakamura and Steinsson (2018) policy-news surprise, measured in narrow windows around FOMC announcements (253 announcements, 1994–2026). Its impact on the two-year yield and, when included, on a market-based financial conditions index (FCI) is read off the window, where macroeconomic variables do not move. A daily model carries the financial variables forward from the announcement day, and monthly unemployment and core inflation respond to monthly averages of the daily variables, as in a standard monthly VAR. The model respects the timing of each announcement within its month, which addresses the temporal-aggregation problem stressed by Kilian (2024) and Chudik and Georgiadis (2022); related mixed-frequency approaches include Francis et al. (2011), Ghysels (2016) and Gazzani and Vicondoa (2020). The FCI and the sequence of steps—nominal, real, and relative to r^*—are taken from a companion note on financial conditions (FCI note, 2026) and were fixed before this analysis.

The evidence builds in four steps, each including a quadratic time trend in the monthly equations, which absorbs slow-moving low-frequency movements common to all specifications. With the nominal two-year yield as the only channel, the post-1994 puzzles appear: the unemployment response is unstable across lag lengths and significantly negative with 12 lags, prices barely move, and industrial production rises. Adding the FCI turns the unemployment response conventional, large and significant, but prices fall only weakly and industrial production does not fall. Measuring rates in real terms changes little. Measuring rates and the FCI relative to r^* completes the picture: unemployment rises 0.36–0.40 percentage point at 12 months, the core price level falls 0.53–0.72 percent at 24 months, and industrial production falls about 1 percent at 12 months, all significant at the 90 percent level for every lag length. The magnitudes are close to the benchmark of Bauer and Swanson (2023a): the same for prices, somewhat smaller for industrial production, and larger and more persistent for unemployment.

The robustness exercises sharpen what drives this. Financial conditions are essential for unemployment: without the FCI, no adjustment of the policy rate gives a robust unemployment response. Measuring conditions relative to a neutral rate is essential for prices and output: the price and industrial-production responses are strong whenever the FCI is measured relative to r^*, whether the two-year yield is adjusted with the Laubach–Williams or Kiley (2020) r^*, a market-based forward rate or an 8-year moving average. The results are robust to replacing the Nakamura–Steinsson surprise with the orthogonalized surprise of Bauer and Swanson (2023a) or the pure-policy component of Jarociński and Karadi (2020), which give slightly larger responses, and to re-estimating the daily dynamics in the bootstrap. The unemployment response is robust to the treatment of the pandemic, and is larger in a sample that ends in 2019; the significant price response, by contrast, depends on data since 2020.

These results are complementary to the information-effect and predictability literatures. Our propagation is estimated from the reduced-form link between financial conditions and the economy, which we argue makes it less exposed to the correlation between surprises and prior news emphasized by Bauer and Swanson (2023a); and the surprise reaches macro variables only through financial conditions, which screens out information channels that bypass markets (information that moves equity prices in the window is still included). Combining orthogonalized or decomposed surprises with propagation through financial conditions relative to r^* is a natural next step. To our knowledge, propagating high-frequency surprises through a broad financial conditions index

measured relative to r^* has not been examined.

2 Data and method

Surprises and window responses. The shock is the Nakamura–Steinsson policy-news factor from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025): the first principal component of changes in five short-term interest-rate futures around 253 FOMC announcements, February 1994 to September 2026, in the database's monetary-event window: from 10 minutes before the statement to 60 minutes after the start of the press conference, or the 30-minute statement window when there is no press conference. We scale it so that the two-year Treasury yield rises 1 percentage point in the window. The window response of the FCI is computed from the changes in the 10-year yield, the S&P 500 and the dollar index in the same database, combined with the FCI's weights; it is 0.99 per percentage point, close to one by construction, since the weights are scaled so that a unit of the index corresponds to a policy move that raises the two-year yield 1 percentage point. Unemployment and prices are zero in the window.

Financial conditions. The FCI is the market index of FCI note (2026): a weighted sum of the 10-year Treasury yield, equity valuation (cyclically adjusted price–earnings ratio, CAPE) and the real broad dollar, with weights proportional to their responses to the same policy news. It comes in three versions, built in the same steps as in that note. *Nominal*: the variables as measured. *Real*: the 10-year yield net of 10-year expected inflation. r^**-gap* (baseline): the real 10-year yield net of the one-sided Laubach–Williams r^* (New York Fed, current estimates), and equities as the gap between the earnings yield and r^*. The two-year yield follows the same rule: nominal; net of two-year expected inflation (Cleveland Fed; Haubrich et al., 2012); and net of expected inflation and r^*. Daily versions follow the daily 10-year yield, CAPE and dollar and are shifted, smoothly within each month, so that their monthly averages equal the monthly indexes exactly; the adjustments for expected inflation and r^* move slowly, so the window responses are the same in every version.

Model. The model has a daily and a monthly block. In the daily block, the two-year yield and (when included) the FCI follow a heterogeneous autoregression in their 1-, 5-, 22- and 66-day averages and respond to the current and previous month's unemployment gap and core inflation (estimated 1990–2026). In the monthly block, the unemployment gap (unemployment rate less the CBO natural rate) and core PCE inflation are rows of a monthly VAR with L lags of each and L lags of the monthly averages of the daily financial variables, estimated by OLS over February 1994–September 2026 with March–December 2020 excluded as dependent observations; L is 3, 6 or 12. Every monthly equation includes a constant and a quadratic time trend. The trend absorbs slow-moving common movements in the levels of unemployment, inflation and rates over the three decades; it is deterministic and does not enter the impulse responses directly. Within each month macro conditions are set first, and markets respond to them; in the announcement month, macro variables have not yet moved. The response to a surprise is computed by placing the window impact on the announcement day (drawn from the empirical distribution of FOMC dates within the month), simulating the daily block forward, averaging within months, and iterating the monthly block. This is a restricted mixed-frequency VAR (Ghysels, 2016) with equal weights within the month, the restriction implied by a standard monthly VAR. A test of equal within-month weights does not reject it for unemployment; it is rejected for core inflation, although the gain in fit is very small.

Identifying assumption. The surprise affects unemployment and prices only through the variables in the daily block, and the economy responds to a given change in those variables in the same way whatever caused it. The first part is weaker the more complete the set of financial

variables; the second is the assumption behind any use of financial conditions indexes to summarize the stance of policy. It is strong: equity prices, for example, partly anticipate activity. A direct mixed-frequency regression of unemployment on the surprises that does not impose the assumption (Chudik and Georgiadis, 2022; Chudik and Kilian, 2026) gives wrong-signed point estimates, driven largely by the 2007–09 announcements, and is about ten times less precise than our estimates; it cannot discriminate between them and zero.

Inference. We resample announcements (which re-estimates the window impact and the distribution of announcement dates) and residuals of the monthly equations, 200 draws; the daily block, the FCI weights and r^* are held fixed. Section 4 shows that re-estimating the daily block in each draw, with 1,000 draws, leaves the intervals essentially unchanged. We report 90 percent intervals, the convention in recent high-frequency studies (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016), and also 68 percent intervals, which Jarociński and Karadi (2020) report alongside 90 percent ones; Gertler and Karadi (2015) use 95 percent.

3 Results

Tables 1 and 2 report the responses of the unemployment gap and the core PCE price level at 12, 24 and 36 months, for 3, 6 and 12 monthly lags, at each step. Figure 1 shows the four key cases.

Table 1: Unemployment gap (percentage points), per 1 percentage point policy-induced rise in the two-year yield

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
Nominal	yes	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Real	no	0.14†	−0.04	−0.20**	0.33**	0.12	−0.13	0.42**	0.23	−0.01
Real	yes	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
r^*-gap	no	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
r^*-gap	yes	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**

Columns: horizon and number of monthly lags. **: 90 percent interval excludes zero; †: 68 percent interval excludes zero (90 percent does not). Rates and FCI at each step as described in Section 2; all specifications include a quadratic time trend. Bauer and Swanson (2023a), benchmark: +0.19 at 12 months (90 percent band excludes zero), +0.17 and +0.15 at 24 and 36 months (band includes zero), read from their Figure 8.

Table 2: Core PCE price level (percent), per 1 percentage point policy-induced rise in the two-year yield

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
Nominal	yes	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
Real	no	−0.14†	−0.08	−0.10	−0.25†	−0.23	−0.16	−0.33†	−0.37†	−0.26
Real	yes	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
r^*-gap	no	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
r^*-gap	yes	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**

See Table 1. Bauer and Swanson (2023a), benchmark for the CPI (not core PCE): −0.36, −0.48 and −0.6 at 12, 24 and 36 months, 90 percent bands exclude zero.

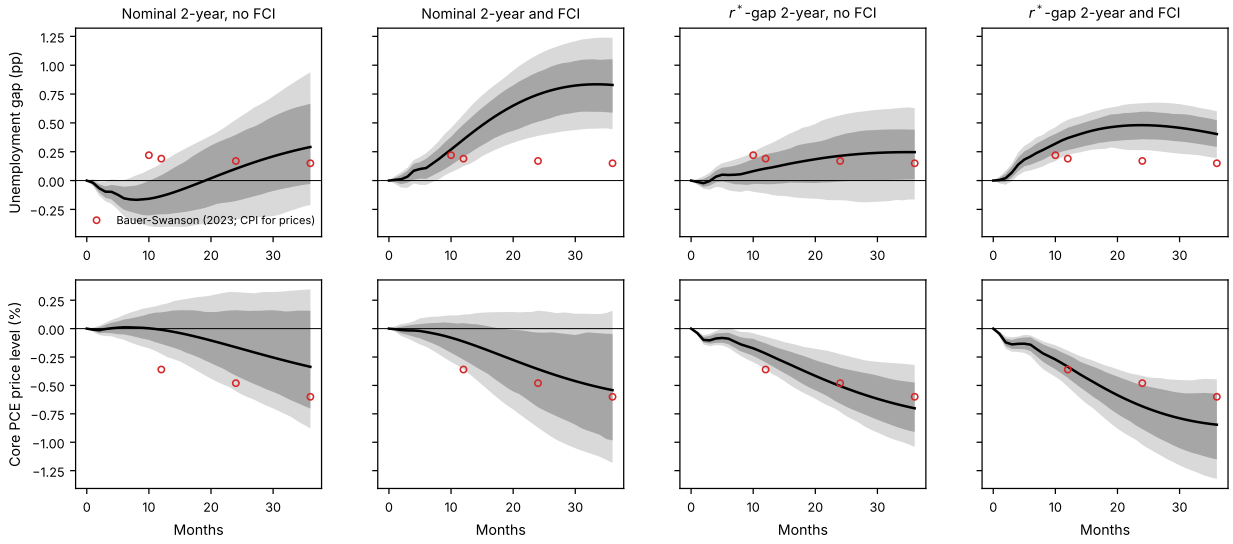


Figure 1: Responses to a policy move that raises the two-year yield 1 percentage point (6 lags). Dark band: 68 percent; light band: 90 percent. Circles: Bauer and Swanson (2023a), read from their Figure 8 (unemployment rate; CPI rather than core PCE).

Step 1: the standard set-up reproduces the puzzles. With the nominal two-year yield as the only channel—the policy indicator of Bauer and Swanson (2023a)—the unemployment response depends on the lag length and is significantly negative with 12 lags (−0.27 at 12 months), and the price level barely moves. This is the post-1994 pattern documented by Ramey (2016) and the one Bauer and Swanson (2023a) obtain with unadjusted surprises. Industrial production rises significantly (Section 4).

Step 2: financial conditions. Adding the FCI turns the unemployment response conventional, large and significant at the 90 percent level at every horizon and lag length: 0.28–0.44 percentage point at 12 months and 0.6–0.8 at 24–36 months. The price level falls by 0.2–0.5 percent at 24–36 months, significant only at the 68 percent level and only with 6 and 12 lags, and industrial production does not fall.

Step 3: real rates. Measuring the two-year yield and the FCI net of expected inflation changes little, as in the companion note, where the nominal and real indexes are close after 1990; the price

decline becomes significant at the 68 percent level at every horizon and lag length.

Step 4: rates relative to r^*. Measuring the two-year yield and the FCI relative to r^* completes the picture. Unemployment rises 0.36–0.40 percentage point at 12 months, 0.47–0.50 at 24 months and 0.35–0.43 at 36 months; the core price level falls 0.33–0.34 percent at 12 months, 0.53–0.72 at 24 months and 0.57–0.97 at 36 months; all are significant at the 90 percent level for every lag length, and nearly invariant to it. The unemployment response is more moderate and less persistent than with the nominal FCI. Without the FCI, the r^*-gap rate gives a strong price response but a weak and lag-sensitive unemployment response (significant only with 3 lags).

Comparison with Bauer and Swanson (2023a). Their benchmark proxy VAR (1973–2020, with surprises from FOMC announcements and Chair speeches, orthogonalized to pre-announcement news) gives an unemployment response of about 0.2 percentage point at 12 months, 0.17 at 24 and 0.15 at 36 months, and a CPI decline of about 0.36, 0.48 and 0.6 percent, per percentage point on the two-year yield. Our step-4 price responses (core PCE) are about the same; our unemployment response is about twice as large and more persistent, with intervals that exclude zero through 36 months where theirs include zero after about a year and a half. Their estimates use 30 years of instrument data, Chair speeches and orthogonalization; ours use FOMC windows alone and no orthogonalization. Earlier versions of our model with Greenbook-purged surprises gave nearly identical results.

4 Robustness

Industrial production. Table 3 adds industrial production to the monthly block. Without the FCI, industrial production rises after a tightening with the nominal and real rates and is insignificant with the r^*-gap rate. With the nominal or real FCI it does not fall significantly. Only at the r^* step does it fall, by 0.84–1.05 percent at 12 months, significant at the 90 percent level for every lag length, against about 1.5 percent in Bauer and Swanson (2023a); it recovers within about three years. Unemployment and price responses are similar to the core system. In exploratory work, adding a commodity price index as well produced rising commodity prices after a tightening, a reminder that the propagation assumption is least plausible for the most forward-looking, demand-sensitive variables.

Table 3: Monthly block with industrial production: unemployment gap (top) and industrial production (bottom)

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	0.07	−0.04	−0.19†	0.26†	0.16	−0.04	0.26	0.30†	0.19
Nominal	yes	0.41**	0.36**	0.22**	0.60**	0.59**	0.47**	0.51**	0.56**	0.49**
Real	no	0.12	0.03	−0.14†	0.24†	0.15	−0.02	0.24†	0.23	0.15
Real	yes	0.41**	0.37**	0.24**	0.54**	0.53**	0.46**	0.43**	0.48**	0.45**
r^*-gap	no	0.16†	0.16†	0.03	0.30**	0.27†	0.14	0.33**	0.30†	0.25†
r^*-gap	yes	0.44**	0.42**	0.34**	0.52**	0.50**	0.50**	0.42**	0.42**	0.42**

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	1.06**	1.26**	1.65**	1.03†	1.26†	1.64**	1.16†	1.00†	1.10†
Nominal	yes	−0.16	−0.29	0.11	−0.09	−0.04	0.15	0.45	0.39	0.43
Real	no	0.64†	0.77**	1.13**	0.72	0.97†	1.28**	0.87†	0.85†	0.91†
Real	yes	−0.33	−0.52	−0.23	−0.11	−0.11	−0.01	0.42	0.31	0.34
r^*-gap	no	0.13	−0.04	0.27	0.13	0.23	0.32	0.31	0.44	0.21
r^*-gap	yes	−0.84**	−1.05**	−0.95**	−0.46†	−0.48	−0.80**	0.18	0.08	−0.21

See Table 1. Bauer and Swanson (2023a), industrial production: about −1.5, −1.0 and −0.85 at 12, 24 and 36 months, read from their Figure 8. Price-level responses (not shown) are as in Table 2.

Which neutral-rate adjustment? Table 4 holds the FCI at the three indexes of the companion note (nominal, real and r^*-gap), keeps the quadratic trend, and varies only how the two-year yield is adjusted: by the Laubach–Williams r^* (baseline), by Kiley (2020)'s r^*, by a market-based measure of the nominal neutral rate—the 9–10-year forward rate from the Federal Reserve's fitted yield curve (Gürkaynak et al., 2007), which replaces both the expected-inflation and r^* adjustments—or by the trailing 8-year average of the two-year yield. Three results stand out. First, financial conditions are essential for unemployment: without the FCI, the unemployment response is weak with the model-based r^* and significantly negative with the forward rate or the moving average; with any of the FCIs, it is positive at every horizon (0.17–0.48 percentage point at 12 months) and significant at the 90 percent level in almost all cases. Second, the price response is strong whenever the FCI is measured relative to r^*: with the r^*-gap FCI, the price level falls 0.3–0.7 percent at 24 months whatever the adjustment of the two-year yield, significant at the 90 percent level for every lag length. With the nominal or real FCI, it is strong when the two-year yield is adjusted with a model-based r^* (Laubach–Williams or Kiley) and weak when it is adjusted with the forward rate (whose term premium fell over the sample) or the moving average. Third, Kiley's r^* gives results very close to Laubach–Williams.

Table 4: Alternative trends in rates: unemployment gap (top) and core PCE price level (bottom)

		12 months			24 months			36 months	
Rate	FCI	3	6	12	3	6	12	3	6
2-year r^*-gap, Laubach–Williams (baseline)	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†
2-year r^*-gap, Laubach–Williams (baseline)	nominal	0.48**	0.43**	0.35**	0.63**	0.63**	0.58**	0.49**	0.53**
2-year r^*-gap, Laubach–Williams (baseline)	real	0.46**	0.42**	0.36**	0.59**	0.59**	0.57**	0.45**	0.49**
2-year r^*-gap, Laubach–Williams (baseline)	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**
2-year r^*-gap, Kiley	none	0.13†	−0.02	−0.15†	0.26†	0.07	−0.11	0.30**	0.13
2-year r^*-gap, Kiley	nominal	0.44**	0.38**	0.32**	0.58**	0.60**	0.58**	0.43**	0.50**
2-year r^*-gap, Kiley	real	0.44**	0.38**	0.32**	0.55**	0.58**	0.57**	0.40**	0.48**
2-year r^*-gap, Kiley	r^*-gap	0.39**	0.36**	0.33**	0.45**	0.48**	0.52**	0.31**	0.39**
2-year − 9–10y forward rate	none	−0.20**	−0.20†	−0.23**	−0.25**	−0.17	−0.23†	−0.24**	−0.11
2-year − 9–10y forward rate	nominal	0.26**	0.25**	0.18†	0.33**	0.42**	0.35**	0.26**	0.35**
2-year − 9–10y forward rate	real	0.26**	0.26**	0.20†	0.31**	0.40**	0.35†	0.24**	0.33**
2-year − 9–10y forward rate	r^*-gap	0.22**	0.23**	0.26**	0.26**	0.35**	0.40**	0.21**	0.30**
2-year − 8-year average	none	−0.16†	−0.28**	−0.30**	−0.05	−0.18	−0.30†	0.02	−0.06
2-year − 8-year average	nominal	0.30**	0.25**	0.21**	0.39**	0.43**	0.40**	0.27**	0.35**
2-year − 8-year average	real	0.27**	0.21**	0.17†	0.36**	0.39**	0.36**	0.26**	0.33**
2-year − 8-year average	r^*-gap	0.22**	0.17**	0.21**	0.33**	0.36**	0.41**	0.29**	0.37**

		12 months			24 months			36 mo	
Rate	FCI	3	6	12	3	6	12	3	6
2-year r^*-gap, Laubach–Williams (baseline)	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.7
2-year r^*-gap, Laubach–Williams (baseline)	nominal	−0.25**	−0.28**	−0.30**	−0.42**	−0.59**	−0.66**	−0.50**	−0.7
2-year r^*-gap, Laubach–Williams (baseline)	real	−0.26**	−0.29**	−0.33**	−0.44**	−0.59**	−0.70**	−0.52**	−0.7
2-year r^*-gap, Laubach–Williams (baseline)	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.8
2-year r^*-gap, Kiley	none	−0.23**	−0.16†	−0.18**	−0.37**	−0.34†	−0.29†	−0.44**	−0.
2-year r^*-gap, Kiley	nominal	−0.23**	−0.24**	−0.21†	−0.39**	−0.50**	−0.41†	−0.43**	−0.6
2-year r^*-gap, Kiley	real	−0.25**	−0.25**	−0.25**	−0.40**	−0.51**	−0.46†	−0.44**	−0.6
2-year r^*-gap, Kiley	r^*-gap	−0.34**	−0.33**	−0.29**	−0.48**	−0.62**	−0.58**	−0.47**	−0.6
2-year − 9–10y forward rate	none	0.01	−0.04	−0.08	0.04	−0.04	−0.06	0.05	−0.
2-year − 9–10y forward rate	nominal	−0.05	−0.12	−0.14†	−0.07	−0.21	−0.25	−0.06	−0.
2-year − 9–10y forward rate	real	−0.06	−0.14†	−0.18†	−0.07	−0.21	−0.30†	−0.06	−0.
2-year − 9–10y forward rate	r^*-gap	−0.22**	−0.27**	−0.30**	−0.29**	−0.48**	−0.62**	−0.30†	−0.5
2-year − 8-year average	none	−0.08	−0.06	−0.08	−0.25†	−0.27†	−0.23†	−0.37†	−0.
2-year − 8-year average	nominal	−0.13†	−0.15†	−0.16†	−0.27†	−0.38**	−0.39†	−0.31†	−0.4
2-year − 8-year average	real	−0.14†	−0.15†	−0.18†	−0.28†	−0.37**	−0.42†	−0.32†	−0.4
2-year − 8-year average	r^*-gap	−0.25**	−0.24**	−0.24**	−0.46**	−0.61**	−0.63**	−0.56**	−0.8

See Table 1. FCI column: the FCI version used. All specifications include the quadratic trend. The FCI is always one of the companion note's three indexes; rows differ only in the adjustment of the two-year yield. The 8-year average and Laubach–Williams trends are close until 2015 (averages 5.3 and 5.5 percent over 1994–2007; 2.3 and 2.5 over 2008–2015) and diverge after 2016 (1.4 and 3.6).

The pandemic. The baseline drops March–December 2020 as dependent observations, but 2020 values still enter the 2021 equations as lags. Table 5 also drops 2021 and, separately, ends the sample in December 2019 (estimating the daily block through 2019 as well). The unemployment response is robust: with the r^*-gap rate and FCI it is 0.26–0.31 percentage point at 12 months when

2021 is dropped and 0.42–0.44 when the sample ends in 2019, significant at the 90 percent level at almost every horizon and lag length; without the FCI it is insignificant in both samples. The price response depends on the recent period: dropping 2021, the price level still falls significantly with 6 and 12 lags (−0.43 and −0.63 percent at 24 months), but in the sample ending in 2019 it barely moves. The significant price response therefore draws heavily on the period since 2020, when inflation rose and fell with a large swing in the stance of policy relative to r^*.

Table 5: Sample: unemployment gap (top) and core PCE price level (bottom), r^*-gap rate

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Baseline sample	r^*-gap (LW)	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Also drop 2021	r^*-gap (LW)	0.31**	0.26**	0.30**	0.38**	0.33**	0.38**	0.29**	0.28†	0.30**
End in 2019	r^*-gap (LW)	0.42**	0.42**	0.44**	0.45**	0.50**	0.60**	0.34**	0.42**	0.59**
Baseline sample	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
Also drop 2021	none	0.03	−0.03	−0.09	0.09	−0.01	−0.09	0.12	−0.01	−0.11
End in 2019	none	0.05	0.05	0.00	0.09	0.14	0.14	0.11	0.19	0.28

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Baseline sample	r^*-gap (LW)	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Also drop 2021	r^*-gap (LW)	−0.16†	−0.21†	−0.28**	−0.24†	−0.43**	−0.63**	−0.29†	−0.60**	−0.93**
End in 2019	r^*-gap (LW)	−0.02	−0.03	−0.05	0.02	−0.00	−0.12	0.03	−0.03	−0.22†
Baseline sample	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
Also drop 2021	none	−0.11†	−0.09	−0.13†	−0.18†	−0.25†	−0.35**	−0.22†	−0.38†	−0.59**
End in 2019	none	0.02	0.02	0.08†	0.06	0.02	0.03	0.10	0.02	−0.06

See Table 1. "End in 2019" also estimates the daily block through 2019.

Alternative shocks. Table 6 replaces the Nakamura–Steinsson surprise with two alternatives from the literature, keeping everything else unchanged: the orthogonalized surprise of Bauer and Swanson (2023a) (their posted series for FOMC announcements, 201 of our events through 2019), and the "poor man's" sign restriction of Jarociński and Karadi (2020), which keeps only announcements where the two-year yield and the S&P 500 move in opposite directions in the window (157 events), the pure-policy component. Both give slightly larger responses than the baseline: at the r^* step, unemployment rises 0.40–0.45 percentage point at 12 months and the price level falls 0.55–0.77 percent at 24 months, significant at the 90 percent level for every lag length. Both shocks move the FCI somewhat more per point on the two-year yield (1.07 and 1.09 against 0.99), consistent with the removal of information or predictable components that move rates and equities in the same direction. Without the FCI, the shock enters only through its normalized two-year move, so the responses are essentially the same for all three shocks. Our results are therefore not driven by the information or predictable components of the Nakamura–Steinsson surprise; propagation through financial conditions relative to r^* and the remedies of Bauer and Swanson (2023a) and Jarociński and Karadi (2020) work in the same direction.

Table 6: Alternative shocks: unemployment gap (top) and core PCE price level (bottom), with the FCI

		12 months			24 months			36 months		
Shock	Step (with FCI)	3	6	12	3	6	12	3	6	12
Nakamura–Steinsson (baseline)	Nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Nakamura–Steinsson (baseline)	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Bauer–Swanson orthogonalized	Nominal	0.47**	0.39**	0.32**	0.81**	0.79**	0.67**	0.81**	0.87**	0.80**
Bauer–Swanson orthogonalized	r^*-gap	0.44**	0.41**	0.40**	0.52**	0.52**	0.54**	0.40**	0.44**	0.46**
Jarociński–Karadi pure policy	Nominal	0.48**	0.40**	0.33**	0.82**	0.81**	0.69**	0.83**	0.89**	0.81**
Jarociński–Karadi pure policy	r^*-gap	0.45**	0.42**	0.41**	0.53**	0.54**	0.55**	0.41**	0.45**	0.47**

		12 months			24 months			36 months	
Shock	Step (with FCI)	3	6	12	3	6	12	3	6
Nakamura–Steinsson (baseline)	Nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†
Nakamura–Steinsson (baseline)	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**
Bauer–Swanson orthogonalized	Nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.32	−0.34	−0.55†
Bauer–Swanson orthogonalized	r^*-gap	−0.36**	−0.35**	−0.35**	−0.55**	−0.71**	−0.76**	−0.59**	−0.88**
Jarociński–Karadi pure policy	Nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.33	−0.34	−0.56†
Jarociński–Karadi pure policy	r^*-gap	−0.36**	−0.36**	−0.35**	−0.55**	−0.72**	−0.77**	−0.60**	−0.89**

See Table 1. All shocks scaled to raise the two-year yield 1 percentage point in the window; all specifications include the quadratic trend.

Inference. Re-estimating the daily block in every draw (a moving-block bootstrap of the daily regressions with 66-day blocks) and raising the number of draws to 1,000 leaves the intervals essentially unchanged: for the baseline (r^*-gap rate and FCI) every significance marker in Tables 1 and 2 is the same, and only a few markers elsewhere change between the 68 and 90 percent levels (Appendix Table 7). The r^* estimate and the FCI weights are held fixed.

Other checks. In exploratory versions of the model, adding a Baa–Treasury spread to the daily block lowered the unemployment response by a third to a half; because no intraday credit series exists, the spread's window response could not be measured cleanly, and we do not pursue it here, but credit is the most important omitted channel (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019). Replacing the model's daily dynamics with the directly estimated post-announcement path of the financial variables for the first one to two months gives larger but much less precise responses.

5 Conclusion

High-frequency surprises identify monetary policy shocks well; the difficulty has been in tracing their effects on the economy. Holding the surprise fixed and changing only its propagation, we find that two standard features of the transmission mechanism account for much of the post-1994 puzzle. Policy works through financial conditions as a whole, and it is the level of rates and financial conditions relative to a slow-moving neutral rate that matters for spending and inflation. Propagating window-identified surprises through a broad financial conditions index, measured relative to r^*, gives conventional, precise and lag-robust responses of unemployment, prices and industrial production, of a size comparable to the best recent estimates.

The two ingredients play different roles. Financial conditions are essential for the unemployment response, in every variant and sample we consider. Measuring financial conditions relative to r^*

is essential for the responses of prices and industrial production, and it matters less how the neutral rate is measured once the FCI is: model-based estimates, a forward rate and a moving average all deliver them. The unemployment response is robust to the pandemic; the price response draws heavily on the period since 2020. These findings complement the information-effect and predictability literatures; the mechanisms can operate together, and combining orthogonalized or decomposed surprises with our propagation is a natural next step. The main limitations are the identifying assumption—that policy reaches the economy through the financial variables in the model, and that the economy responds to them the same way whatever moved them—the omission of credit spreads, and intervals that do not include uncertainty about r^* or the daily dynamics. The broader lesson is that the decline in r^* over the past three decades is not only a fact about the level of rates but also a measurement problem for estimating the effects of policy.

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A Inference with the daily block re-estimated

Table 7: Core steps with the daily block re-estimated in each draw (1,000 draws): unemployment gap (top) and core PCE price level (bottom)

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
Nominal	yes	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Real	no	0.14†	−0.04	−0.20†	0.33**	0.12	−0.13	0.42**	0.23	−0.01
Real	yes	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
r^*-gap	no	0.19**	0.10	−0.03	0.34**	0.21†	0.04	0.38**	0.25†	0.07
r^*-gap	yes	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26	−0.34	−0.18
Nominal	yes	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31	−0.33	−0.54†	−0.49
Real	no	−0.14†	−0.08	−0.10	−0.25†	−0.23†	−0.16	−0.33†	−0.37†	−0.26
Real	yes	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
r^*-gap	no	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
r^*-gap	yes	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**

As Tables 1 and 2, with a moving-block bootstrap (66-day blocks) of the daily regressions in each draw; r^* and the FCI weights held fixed.